

Assignment 1: Learning, Generalization, and Model Choice

Foundations and Practice of Modern AI Systems | Fall 2026

Date Assigned: September 10, 2026 Due Date: 11pm, September 22, 2026

Max Points: 100

Purpose:

The Week 0 lab asked, “*Here is a model. Can you recognize and run a basic machine-learning workflow?*” Building on that foundation, this assignment asks you to begin making **defensible choices about representation, model selection, evaluation, and generalization**. You will independently carry out and interpret a basic machine-learning workflow, examine how modeling choices affect results, and use evidence to justify your decisions. The emphasis is on understanding **why choices matter and how to evaluate them appropriately**, rather than implementing a complex algorithm from scratch. Learning outcomes addressed LO1, LO2, LO6 + LO8.

The scenario is “**Screening Wine Quality with K-Nearest Neighbors**”

Read and Follow Assignment Instructions Carefully:

- This is an individual assignment. All work submitted must be your own.
- Use of AI tools is **NOT** permitted. The purpose of Assignment 1 is to establish a baseline of your own understanding and ability to formulate problems, reason through the methods, and implement them independently before AI-assisted work is introduced later in the course.
- Deliverable: One reproducible **Jupyter notebook with code, figures/tables, and concise responses**. The notebook should be named `<lastname>_<firstname>_assignment1.ipynb` and uploaded to Brightspace.
- Make sure you copy each question with the question number as a Markdown Cell in your Jupyter notebook and have the code response right below it. Points will be deducted if it is difficult to locate the question and response.
- Make sure you comment your code. Points will be deducted if code logic is not apparent.
- This assignment should take about 4-5 hours total.

Dataset:

Use the UCI Wine Quality dataset (white wine), provided as a cleaned course copy. Each row contains physicochemical measurements. The target is binary: $\text{quality} \leq 5$ = lower quality (0); $\text{quality} > 5$ = higher quality (1).

- Use a 80-20 split (80 training, 20 test)
- Do not use the held-out test set for model selection.
- You may use NumPy, Pandas, Matplotlib, and scikit-learn, including its KNN implementation.

Part A — Understand the Problem and Data [15 points]

A1. [4 points] State the prediction task. What are the inputs, target, and learning setting? Why is this classification?

A2. [5 points] Report the number of examples, input features, class distribution, and ranges of at least four features. Identify one feature-scale difference that could matter for KNN.

A3. [6 points] Explain why a held-out test set is needed and why repeatedly using it while choosing k weakens the evaluation.

Part B — Representation and Baseline Model [20 points]

B1. [6 points] Fit KNN with $k=5$, Euclidean distance, and uniform voting using original feature values. Use cross-validation on training data; report mean accuracy and F1.

B2. [6 points] Standardize features using training-data statistics only. Repeat B1 and report the same metrics.

B3. [5 points] Explain why standardization can change KNN predictions, referring explicitly to distance and feature scale.

B4. [3 points] Explain why fitting the scaler on the complete dataset would be data leakage.

Part C — Model Choice and Generalization [30 points]

C1. [10 points] Using standardized features, compare $k \in \{1, 5, 11, 21\}$ with 5-fold cross-validation. Use F1 as the primary selection metric and present a concise table.

C2. [8 points] For two informative k values, compare Euclidean and Manhattan distance. Does the preferred distance change your conclusion? Support your answer with evidence.

C3. [8 points] Choose a final configuration. Explain the trade-off between very small and larger k in terms of sensitivity, decision boundaries, and generalization.

C4. [4 points] Before testing, predict in one sentence how the selected model will perform relative to its cross-validation estimate.

Part D — Final Evaluation [20 points]

D1. [8 points] Evaluate the selected model once on the held-out test set. Report accuracy, precision, recall, F1, and a confusion matrix.

D2. [5 points] Interpret false positives and false negatives for the distributor's screening task. Which error is more consequential and why?

D3. [4 points] Compare test performance with cross-validation. Is the difference evidence of a generalization problem? Explain.

D4. [3 points] State one conclusion the test result supports and one it does not support.

Part E — Evidence-Based Recommendation [15 points]

E1. [10 points] Write a 250–350 word recommendation stating whether the model is useful as a screening tool, citing two pieces of evidence and at least two limitations.

E2. [5 points] Name one additional experiment or data source you would want before deployment and explain what uncertainty it addresses.